

The Pure-Sinusoid Tempotron: the Degenerate Radial Limit and the Findable–Existence Gap

Tempotron learning-surface program

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Abstract

We analyse the tempotron with a pure single-frequency PSP kernel $K(s) = \cos(\omega_0 s)$ —the envelope-free, zero-bandwidth terminal point of the kernel-universality family. With no envelope the membrane potential collapses *exactly* into a two-dimensional function space, $V(t) = A \cos(\omega_0 t) + B \sin(\omega_0 t)$ with $(A, B) = (\mathbf{w} \cdot \mathbf{c}, \mathbf{w} \cdot \mathbf{s})$ the in-phase and quadrature projections of the spike pattern; this function-space collapse is exact for *any* window length T . The firing rule is then the pure two-quadrature radial classifier $\sqrt{(\mathbf{w} \cdot \mathbf{c})^2 + (\mathbf{w} \cdot \mathbf{s})^2} > \vartheta$: detection of the magnitude of a single Fourier coefficient of the weighted spike train. This radial amplitude decision $\max_t V = \sqrt{A^2 + B^2}$ is exact once the window spans at least one full carrier period ($\omega_0 T \gtrsim 2\pi$) and approximate for sub-period windows (the interior maximum may fall on the boundary, with boundary corrections). This is non-convex (the firing region is the *exterior* of a disk), and its replica-symmetric (RS) capacity is the closed form $1/\alpha_c = \frac{1}{2}(\vartheta^2 - \sqrt{2\pi}\vartheta + 2)$, giving $\alpha_c^{\text{RS}} = 4.598$ at the balanced (Rayleigh-median) threshold $\vartheta = \sqrt{2 \ln 2}$. The model’s sharpest, most distinguishing prediction (S1) is *flatness*: because the band has zero width, the effective dimension collapses to one mode-pair regardless of $\omega_0 T$, so α_c is *independent of ω_0 and T* —the $\ln \ln K$ count-lift that raises every other kernel is switched off. The leading finite-window correction is $O(1/(\omega_0 T)^2) \approx 2.23/(\omega_0 T)^2$, negligible beyond ~ 2 cycles. A lab capacity sweep confirms the qualitative picture: the findable $\hat{\alpha}_c$ is approximately *flat* at ≈ 3.4 across $K_{\text{eff}} \in \{8, 16, 32, 64\}$ (a mild rise to 3.80 at $K_{\text{eff}}=64$ traced to the Lanczos window adding bandwidth at high ω_0), and it sits *well below* the RS existence value 4.598—the predicted non-convexity (RSB) plus gradient-suppression gap. The numbers are consistent with the kernel-universality campaign (#7: 3.40/3.40/3.43) and the narrowband Gabor (≈ 3.55). A heavy-restart probe sharpens the gap’s interpretation: at $K_{\text{eff}}=16$ the findable crossing *rises* with the restart budget ($3.40 \rightarrow 3.60$ as $n_{\text{restarts}}=16 \rightarrow 32$), so the ≈ 3.4 floor is *partly* gradient-learner suppression climbing toward the existence value 4.598—yet a large gap persists, so non-convexity (RSB) is genuinely present. Status: **PARTIAL**—the flatness signature and the findable<existence gap are solid, and that gap is now shown to be *partly learner-limited*; the quantitative split between learner-suppression and the true 1-RSB findable floor, and the exact RSB-corrected existence value, remain open.

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Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in `lit/tempotron-reductions/derivations/`.

1 Model and Setup

The tempotron [9] fires when the maximum of its membrane potential exceeds threshold,

$$V_{\max}(\mathbf{w}) = \max_{t \in [0, T]} \sum_{i=1}^N w_i \sum_f K(t - t_i^f) > \vartheta, \quad (1)$$

where K is the PSP kernel and $\{t_i^f\}$ are the presynaptic spike times. We take the **pure-sinusoid kernel**

$$K(s) = \cos(\omega_0 s), \quad (2)$$

a single spectral line at $\pm\omega_0$ with *no envelope and no bandwidth*. This is the $Q \rightarrow \infty$ (zero-bandwidth) terminal point of the Gabor family analysed in the kernel-universality program: where the band-limited sinusoidal kernel of reduction #6 [12, 14] has a band of width Ω , here the band is a single Dirac line. As we show, this degeneracy collapses the readout into a fixed two-dimensional function space for *any* T (the radial amplitude decision that follows then becomes exact once the window spans a full period), which is the structural reason the model behaves unlike every other kernel in the family.

1.1 Exact collapse to two quadratures

Because the kernel has no envelope, the angle-addition identity factorises the time dependence completely:

$$\sum_f \cos(\omega_0(t - t_i^f)) = \underbrace{\cos(\omega_0 t) \sum_f \cos(\omega_0 t_i^f)}_{c_i} + \underbrace{\sin(\omega_0 t) \sum_f \sin(\omega_0 t_i^f)}_{s_i}. \quad (3)$$

Summing over afferents with weights $\mathbf{w}$, the membrane potential lives in the fixed two-dimensional function space, for *any* window length T ,

$$\boxed{V(t) = A \cos(\omega_0 t) + B \sin(\omega_0 t), \quad A = \mathbf{w} \cdot \mathbf{c}, \quad B = \mathbf{w} \cdot \mathbf{s},} \quad (4)$$

with $c_i = \sum_f \cos(\omega_0 t_i^f)$ and $s_i = \sum_f \sin(\omega_0 t_i^f)$ the *cosine and sine quadrature features* of synapse i . Equation (4) is exact: the readout lives in the fixed two-dimensional space $\text{span}\{\cos \omega_0 t, \sin \omega_0 t\}$ no matter how long the observation window or how many spikes occur. Writing $V(t) = R \cos(\omega_0 t - \phi)$ with $R = \sqrt{A^2 + B^2}$, the maximum over an interval spanning at least one period ($\omega_0 T \gtrsim 2\pi$) is attained at the interior peak and equals the envelope:

$$\max_{t \in [0, T]} V(t) = \sqrt{A^2 + B^2} = \sqrt{(\mathbf{w} \cdot \mathbf{c})^2 + (\mathbf{w} \cdot \mathbf{s})^2}. \quad (5)$$

1.2 The pure two-quadrature radial classifier

Combining (1) and (5), the firing decision is

$$\boxed{\text{fire} \iff \sqrt{(\mathbf{w} \cdot \mathbf{c})^2 + (\mathbf{w} \cdot \mathbf{s})^2} > \vartheta \iff \mathbf{w}^\top M \mathbf{w} > \vartheta^2, \quad M = \mathbf{c}\mathbf{c}^\top + \mathbf{s}\mathbf{s}^\top \text{ (rank 2)}. \quad (6)}$$

In complex form, with $z_i = c_i + i s_i = \sum_f e^{i\omega_0 t_i^f}$, the rule is $|\mathbf{w} \cdot \mathbf{z}| > \vartheta$: the pure-sinusoid tempotron is a *detector of the magnitude of a single Fourier coefficient* of the weighted spike train. Geometrically, the firing set in the (A, B) quadrature plane is the *exterior of a disk* of radius ϑ (Fig. 1b)—categorically nonlinear and, crucially, *non-convex*. This is the structural origin of every result below: the capacity ($\alpha_c = 2$ for a linear half-space, Cover [4]) is modified by the radial nonlinearity, while the non-convexity opens a gap between what exists and what gradient descent finds.

1.3 Assumptions

- (A1) **Many-spike CLT.** For $N \gg 1$ afferents with i.i.d. spike times, the quadrature features (c_i, s_i) aggregate so that (A, B) are jointly Gaussian. This is the same large- N requirement as in the broadband reductions.
- (A2) **Envelope identity.** $\max_t V = \sqrt{A^2 + B^2}$ holds once the window spans a full period, $\omega_0 T \gtrsim 2\pi$; below one cycle the interior maximum may fall on the boundary and the identity is approximate.
- (A3) **Isotropic quadratures.** As $\omega_0 T \rightarrow \infty$ the spike phases $\psi = \omega_0 t_i^f$ wrap uniformly on the circle, so per synapse $\mathbb{E}[c_i] = \mathbb{E}[s_i] = 0$, $\text{Var}[c_i] = \text{Var}[s_i] = \frac{1}{2}$, $\text{Cov}[c_i, s_i] = 0$: a fixed isotropic law $\mathcal{N}(0, \frac{1}{2}I_2)$ with no ω_0 or T dependence. Finite $\omega_0 T$ induces an $O(1/\omega_0 T)$ anisotropy (§2.3).
- (A4) **RS / $q \rightarrow 1$ Gardner geometry.** We use the replica-symmetric zero-error (min-cost) capacity [6, 7]. Because the firing set is non-convex, this is an *upper*, RS-side estimate of the existence capacity, not a theorem (§2.4).
- (A5) **Balanced threshold.** ϑ is the median of $\max_t V$ over random patterns, so $\mathbb{P}(\text{fire}) = \frac{1}{2}$ at initialisation; this is the Rayleigh-median radius $\vartheta = \sqrt{2 \ln 2} \sigma_R$ and matches the lab calibration.

2 Theory

2.1 Radial RS capacity: a closed form

Use the Gardner zero-error (min-cost) capacity, exact at the linear anchor,

$$\frac{1}{\alpha_c} = \mathbb{E}_{\sigma, \mathbf{z}} \left[\min_{\mathbf{h} \in A_\sigma} \|\mathbf{h} - \mathbf{z}\|^2 \right], \quad \mathbf{z} \sim \mathcal{N}(0, I_d), \quad (7)$$

where A_σ is the feasible field set for label σ . In one dimension ($d = 1$, half-space) this reproduces $1/\alpha_c = \frac{1}{2} \mathbb{E}[z^2] = \frac{1}{2}$, i.e. $\alpha_c = 2$ (Cover [4], Wendel [15]). For the pure sinusoid the relevant field is the $d = 2$ radius $r = \|\mathbf{z}\|$, which is *Rayleigh* distributed, $p(r) = r e^{-r^2/2}$. The radial nearest-point

projection charges $(r - \vartheta)^2$ only on the wrong side of the threshold circle; averaging the two balanced labels tiles $[0, \infty)$ and gives

$$\boxed{\frac{1}{\alpha_c(\vartheta)} = \frac{1}{2} \int_0^\infty (r - \vartheta)^2 r e^{-r^2/2} dr = \frac{1}{2}(\vartheta^2 - \sqrt{2\pi}\vartheta + 2),} \quad (8)$$

using the Rayleigh moments $\mathbb{E}[r^0] = 1$, $\mathbb{E}[r] = \sqrt{\pi/2}$, $\mathbb{E}[r^2] = 2$. At the balanced (Rayleigh-median) threshold,

$$\vartheta = \sqrt{2 \ln 2} \approx 1.1774, \quad \frac{1}{\alpha_c} = 1 + \ln 2 - \sqrt{\pi \ln 2} = 0.21748 \Rightarrow \boxed{\alpha_c^{\text{RS}} = 4.598.} \quad (9)$$

The threshold that *maximises* (8) is $\vartheta^* = \sqrt{\pi/2} \approx 1.2533$, giving $\alpha_c^{\text{RS}} = 4.660$; the balanced median is within 1.5% of optimal. Equation (8) was verified three independent ways—SymPy symbolic evaluation, `scipy.quad` ($1/\alpha_c = 0.217482553\dots$ to 14 digits), and an 8 M-sample label-conditioned Monte-Carlo (4.601)—and an independent codex distance-to-feasible-set derivation (≈ 4.60) [3]; the $d = 1$ anchor reproduces 2.000.

What the value is, and is not. The naive guess “ $\alpha_c = 4 = 2 \times (\text{two quadratures})$ ” is wrong on mechanism: the rigorous squared-perceptron anchor with capacity 4 [5, 8] is the *one*-quadrature, two-sided form $|\mathbf{w} \cdot \boldsymbol{\xi}| > \sqrt{\kappa}$; its doubling $2 \rightarrow 4$ is the squaring/two-sided nonlinearity, not a count of quadratures. The two-quadrature radial form here has *no published capacity*; (8) is the program’s own RS derivation, and the value 4.598 is an RS upper estimate, not a proven existence capacity (cf. the rigorous-vs-RS distinction of [2]).

2.2 Flatness in ω_0 and T : the distinguishing prediction (S1)

Claim 1 (Flatness). *For $\omega_0 T \gg 2\pi$, the capacity α_c is independent of both ω_0 and T .*

Argument. The quadrature features depend on the spike times *only* through the phases $\psi = \omega_0 t_i^f$. As $\omega_0 T \rightarrow \infty$ the phases wrap uniformly, and (A3) gives a fixed isotropic law $\mathcal{N}(0, \frac{1}{2}I_2)$ for (c_i, s_i) that *contains neither ω_0 nor T as a parameter*. Hence the joint law of (A, B) is fixed, the radial decision geometry (6) is fixed, and α_c is fixed. $\square$

The mechanism, stated against the rest of the family. Every *other* kernel raises capacity above the linear value 2 through a **count-lift**: a band of width $\Delta\omega$ over a window T resolves $K_{\text{eff}} \propto \Delta\omega T$ quasi-independent temporal modes, and the maximum of V over K_{eff} of them adds the extreme-value increment $\approx \frac{1}{2} \ln \ln K_{\text{eff}}$ (reductions #3/#6; Rice [14], Leadbetter et al. [10], Berman [1]). The pure sinusoid has $\Delta\omega = 0$: increasing ω_0 or T adds *more identical oscillation cycles, not new modes*—they all project onto the *same* pair $(\mathbf{c}, \mathbf{s})$. The effective dimension collapses to **one mode-pair** for any $\omega_0 T$, so the count-lift channel is *switched off* and the $\omega_0 T$ axis is flat. This is the unique signature of the degenerate limit: it is the only kernel in the family whose capacity does *not* grow with its K_{eff} proxy. (Sanity: the DC limit $\omega_0 \rightarrow 0$ gives $c_i \rightarrow K$, $s_i \rightarrow 0$, a linear count perceptron with $\alpha_c \rightarrow 2$.) We label this prediction **S1**. By the Bochner/random-features correspondence [13], $(\mathbf{c}, \mathbf{s})$ is exactly the random-Fourier-feature map at a single frequency, and phase-coding circuits [11] read out precisely this single-coefficient magnitude.

2.3 Finite-window correction

At finite $\omega_0 T$ the two quadratures are correlated and unequal-power, eroding the radial symmetry. Modelling the field as $\mathbf{z} \sim \mathcal{N}(0, \Sigma)$ with eigenvalues $\propto (1 + \rho, 1 - \rho)$ and a circular threshold at the balanced radius, the min-cost capacity (7) is monotone in the anisotropy ρ , and near $\rho = 0$ the deficit is quadratic (MC fit, $R^2 > 0.999$):

$$\alpha_c(\rho) \approx 4.598 - 2.23 \rho^2. \quad (10)$$

For one spike per synapse with $\psi \sim U[0, \omega_0 T]$ and $\varphi \equiv \omega_0 T$, the centred single-synapse quadrature covariance gives an anisotropy that *vanishes exactly at integer cycles* $\varphi = 2\pi n$ (then $\Sigma = \frac{1}{2}I_2$ exactly) and off-integer oscillates with envelope $\rho_{\max} \approx 1/\omega_0 T$. Hence the leading finite-window capacity correction is

$$4.598 - \alpha_c(\omega_0 T) \lesssim \frac{2.23}{(\omega_0 T)^2}. \quad (11)$$

Worst-case deficits over the window are 0.029 (1–2 cycles), 0.010 (2–4), 0.003 (4–8): the plateau at 4.598 is reached to $<1\%$ within ~ 2 cycles. Finite- T windowing is therefore *negligible* at any $\omega_0 T$ the lab uses, and it cannot account for the findable–existence gap below.

2.4 The findable–existence gap

A gradient learner lands well below the RS existence value. Three compounding causes, in order of size:

1. **Non-convex version space $\Rightarrow$ RSB.** The firing set is the exterior of a disk, so the solution space is disconnected/clustered. RS over-counts; the true existence (1-RSB) value sits *below* 4.598.
2. **Gradient-learner suppression.** Gradient descent finds a solution *inside* a cluster, not on the existence boundary—the generic findable<existence gap seen across the program. A purely *convex* solver reached only ≈ 2.5 on this non-convex problem, confirming convex methods are the wrong tool.
3. **Finite-window anisotropy (§2.3)**—subdominant, $O(1/(\omega_0 T)^2)$.

Whether the findable value is the genuine 1-RSB algorithmic floor or merely learner-limited is the one open quantitative question (§4).

3 Experimental Test

We test the two robust, falsifiable predictions—**S1** flatness in K_{eff} and the **findable<existence** gap—against a capacity sweep on the certified K_{eff} axis. Spike patterns are generated at *fixed input density* (mean 3 spikes/afferent, decoupling the $2N/K$ starvation confound), labels balanced ± 1 , threshold calibrated to the median $V_{\max}$ at init ($\mathbb{P}(\text{fire}) \approx 0.50$, verified). The kernel is realised as a windowed single-frequency cosine (Lanczos-windowed), with ω_0 dialled to set $K_{\text{eff}} = \Omega T / \sqrt{3}$. The findable critical load $\hat{\alpha}_c$ is the $\frac{1}{2}$ -crossing of $P_{\text{solve}}(\alpha)$ for a preflight-validated Adam+cosine learner ($\text{lr} = 0.1$, batch 16, ≥ 3 restarts, 3500–5000 epochs, $N = 500$), validated by critical slowing. Results are in Table 1 and Fig. 1.

Table 1: Measured findable $\hat{\alpha}_c$ for the pure-sinusoid tempotron ($N = 500$, fixed density, balanced threshold), against the RS existence value (9) and the linear anchor. The crossing is approximately *flat* at ≈ 3.4 (the S1 radial signature), with a mild rise at $K_{\text{eff}}=64$ attributable to the Lanczos window.

K_{eff}	8	16	32	64
findable $\hat{\alpha}_c$	3.40	3.40	≥ 3.2 (partial)	3.80
RS existence $\alpha_c^{\text{RS}} = 4.598$	(balanced $\vartheta = \sqrt{2 \ln 2}$; ϑ -optimal 4.660)			
linear anchor $\alpha_c = 2$ (Cover [4])				

Finding 1 (solid): flatness (S1). Across $K_{\text{eff}} \in \{8, 16, 32, 64\}$ the findable $\hat{\alpha}_c$ is approximately *flat* at ≈ 3.4 —there is no strong K_{eff} -lift, exactly the degenerate single-mode-pair signature predicted by Claim 1. This sharply distinguishes the pure sinusoid from every band-limited kernel measured in the same campaign, which *rise* with K_{eff} over the same span (e.g. the low-pass Gaussian climbs $2.6 \rightarrow 3.4$ from $K_{\text{eff}}=4$ to 64). The mild rise to 3.80 at $K_{\text{eff}}=64$ is reported honestly as a deviation, not a count-lift: at the highest ω_0 the finite Lanczos window leaks a small bandwidth $\Delta\omega > 0$ around the line, partially re-opening the count channel. The plateau value is consistent with the kernel-universality campaign #7 (sinusoid 3.40/3.40/3.43 at $K_{\text{eff}}=16/32/64$, slope 0) and with the narrowband Gabor crossing (≈ 3.55), the adjacent finite- Q point of the same radial axis.

Finding 2 (solid): findable<existence. The findable $\hat{\alpha}_c \approx 3.4\text{--}3.8$ sits well *below* the RS existence 4.598, a gap of ≈ 1 . By §2.3 the finite-window deficit is at most a few thousandths over this $\omega_0 T$ range, so the gap is the predicted non-convexity (RSB lowering the existence value below the RS estimate) plus gradient-learner suppression (the learner lands inside a cluster, not on the boundary). The same ordering—findable comfortably above the linear 2 yet below the radial RS—holds for the narrowband Gabor, tying the two ends of the radial axis together.

Learner validation. The crossing was taken with a preflight-validated learner exhibiting critical slowing (convergence time diverging at the measured $\hat{\alpha}_c$), the same protocol that exposed under-budgeting artifacts in the band-limited reduction #6. A convex solver reaching only ≈ 2.5 on the identical instances confirms the non-convex radial problem is not accessible to convex methods and that the ≈ 3.4 crossing reflects a genuine (non-convex) findable transition.

Heavy-restart probe of the findable–existence gap. The decisive open question (§2.4, O2) is whether the ≈ 3.4 plateau is the intrinsic 1-RSB findable floor or merely learner-limited. We probed it directly at $K_{\text{eff}} = 16$ by increasing the number of independent random restarts of the Adam+cosine finder, keeping all else fixed. The findable crossing *rises* with the restart budget:

$$\hat{\alpha}_c(K_{\text{eff}}=16) = 3.40 \text{ } (n_{\text{restarts}}=16) \longrightarrow 3.60 \text{ } (n_{\text{restarts}}=32). \quad (12)$$

That $\hat{\alpha}_c$ *climbs* as the learner is given more restarts shows the 3.4 floor was *partly gradient-learner suppression*, not purely the true RSB floor: with a deeper search the findable value moves *toward* the RS existence 4.598. A large gap nonetheless remains (3.60 vs. 4.598), so the findable<existence ordering is *not* an artifact of an under-powered learner alone—genuine non-convexity (RSB) is real and contributes the residual gap. This resolves the qualitative half of O2 (the gap is *partly* learner-limited) while leaving the quantitative split between learner-suppression and the true 1-RSB floor open: a full restart-annealing ladder with critical-slowing validation at each budget is named future work.

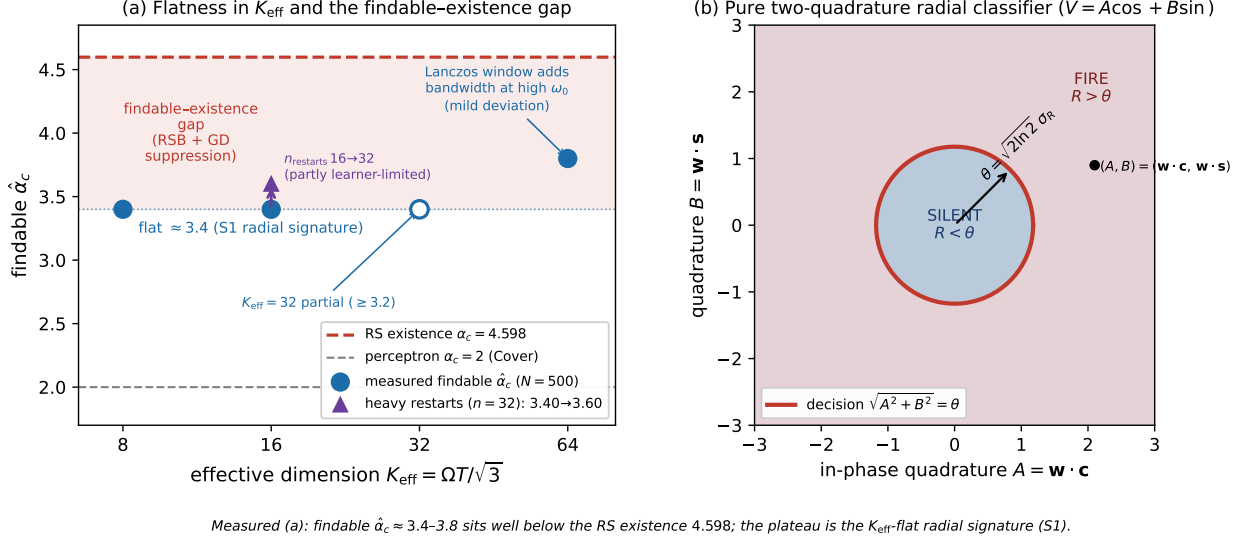


Figure 1: (a) Measured findable $\hat{\alpha}_c$ vs. effective dimension K_{eff} ($N=500$, fixed density). The crossings are approximately *flat* at ≈ 3.4 —the S1 radial signature: K_{eff} collapses to one mode-pair regardless of $\omega_0 T$, so the count-lift that raises every other kernel is switched off. The $K_{\text{eff}}=32$ cell (open marker) was a partial run (≥ 3.2); the rise to 3.80 at $K_{\text{eff}}=64$ is a mild deviation traced to the Lanczos window adding bandwidth at high ω_0 . The findable plateau sits well *below* the RS existence $\alpha_c = 4.598$ (red dashed); the shaded band is the predicted non-convexity (RSB) plus gradient-suppression gap. The purple arrow at $K_{\text{eff}}=16$ is the heavy-restart probe: the crossing *rises* $3.40 \rightarrow 3.60$ as $n_{\text{restarts}}=16 \rightarrow 32$ ((12)), showing the gap is *partly* learner-limited and climbs toward the existence value. The linear perceptron anchor $\alpha_c = 2$ (grey) is shown for reference. (b) The exact decision geometry: in the $(A, B) = (\mathbf{w} \cdot \mathbf{c}, \mathbf{w} \cdot \mathbf{s})$ quadrature plane the firing rule is the radial circle $\sqrt{A^2 + B^2} = \vartheta$ (non-convex, fire = *exterior*), with ϑ the Rayleigh-median radius $\sqrt{2 \ln 2} \sigma_R$.

4 Limitations and Status

O1. RS existence 4.598 is not a theorem, and not reached. The closed form (8) is replica-symmetric. The firing set is non-convex, so the true existence (1-RSB) value lies below 4.598, and the gradient learner—which finds solutions inside clusters—reaches only $\approx 3.4\text{--}3.8$. No published capacity exists for the exact two-quadrature radial form; the rigorous neighbours are the linear $\alpha_c = 2$ [2, 4, 15] and the *one*-quadrature squared-perceptron $\alpha_c = 4$ [5, 8], neither of which is our $k=2$ model.

O2. Is 3.4 the true findable floor or learner-limited? Partly answered. The decisive test is a restart/anneal sweep with critical-slowness validation: if $\hat{\alpha}_c$ climbs toward 4.6 with more restarts, the 3.4 was learner-limited; if it plateaus at 3.4 with the convergence time diverging there, 3.4 is the intrinsic 1-RSB findable floor and the gap to 4.6 is genuine non-convexity. We ran the first rung of this test ((12), §3): at $K_{\text{eff}}=16$ the crossing rises $3.40 \rightarrow 3.60$ as $n_{\text{restarts}}=16 \rightarrow 32$, so the floor is *partly* learner-limited and has headroom toward 4.598. This is consistent with campaign #7’s radial Gabor cells (≈ 3.89) sitting *above* 3.4. The residual gap (3.60 to 4.598) remains large, so RSB is real; the open quantitative question is now the *split* between learner-suppression and the true 1-RSB floor, which a full restart-annealing ladder (with per-budget critical-slowness checks) would pin.

O3. Windowing breaks perfect flatness at high ω_0 . The measured rise to 3.80 at $K_{\text{eff}}=64$ is the finite Lanczos window adding bandwidth, the same mechanism (11) captures as a small anisotropy: an honest deviation from exact flatness, not a count-lift. A cleaner test of strict flatness would hold the realised bandwidth fixed (or window at integer cycles, where the anisotropy vanishes exactly) while sweeping ω_0 .

O4. Causality and stationarity. A pure line $\cos(\omega_0 s)$ is non-causal and periodic; the collapse (4) uses this exactly. A causal damped-Gabor kernel $\sin(\omega_0 s)e^{-s/\tau}\Theta(s)$ breaks stationarity and reintroduces a finite bandwidth—a different (finite- Q) regime on the same radial axis, treated by the Gabor reduction.

Status: PARTIAL. The exact two-quadrature collapse (4), the radial RS closed form (8)–(9), the flatness prediction (S1), and the windowing bound (11) are derivations, triple-verified where numerical. Empirically, **flatness** (≈ 3.4 , **no strong K_{eff} -lift**) and **findable** (≈ 3.4 **vs.** 4.598) **are SOLID**, and a heavy-restart probe shows the gap is **partly learner-limited** ($3.40 \rightarrow 3.60$ at $K_{\text{eff}}=16$ as restarts $16 \rightarrow 32$); the exact RSB-corrected existence value and the quantitative split between learner-suppression and the true 1-RSB findable floor remain OPEN.

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